

F20FO/F21FO - Digital Forensics

Lab 07: Imaging & Data Analysis

Lab 7: Objectives

This lab intends to introduce you to concepts of digital forensic imaging and analysis. At the end of the lab you will be able to perform:

1. Hashing
2. Digital Forensic Imaging and analysis

Notes

There is nothing to submit at the end of this lab, however students are encouraged to log their results in a document. There are some questions **coloured in blue** that you can use as indication to record in your logbook (e.g. lab07_logbook). Often the lab instructions are intentionally open ended, and you will have to figure some things out for yourselves.

The lab will require you to install software and download additional resources from the Resources module for lab07 in Canvas. Changes to your machine should not impact on its functionality - however you are recommended to use a VM.

You can start downloading the resources files while you are reading the lab sheet and save these files to your local machine.

This lab will cover several important fundamental concepts for forensics and security, and will also look at introducing you to hashing and digital forensic imaging.

Lab 7.1: Hashing

As a forensic investigator, you need to ensure that the evidence you are dealing with is not tampered with at all. Even minor changes to evidence files are not permissible. In general computing, you will want to do the same if you are downloading a large file over a network (or the internet). When dealing with numerous files of variable sizes, it is not practical or feasible to manually compare files for changes. Also files can have extremely minor changes that can go completely unnoticed. So how do we ensure that the evidence is intact and has not changed? The answer lies in Hashing. Comparing hash signatures is very important for checking the integrity of a file.

A hash function is a mathematical algorithm that takes an input and produces a fixed length output. The way the algorithm is designed ensures that any change to the original input yields a completely different output string. Hence, hash functions could be employed to

compare the original file to a file later to see if any changes have been made. A hash function is a one-way function - the calculation cannot be reversed.

To put things into perspective, compare the files below.

Paragraph 1

This is a sample test file. We have used this file to demonstrate the effectiveness of hashing to identify the differences in the file. If the content of this file changes the hash value will change too. Hashing is a very valuable principle that lets us verify the integrity of the files we are investigating. It is important that the evidence is not tampered.

Paragraph 1

This is a sample test file. We have used this file to demonstrate the effectiveness of hashing to identify the differences in the file. If the content of this file changes, the hash value will change too. Hashing is a very valuable principle that lets us verify the integrity of the files we are investigating. It is important that the evidence is not tampered.

File1 MD5 hash = bb070ae1452f56921e2571ff1212b314

File 2 MD5 hash = 5c6222bf2be2b1fc3fb0f6c45ee2b1f0

The files are almost identical except for the inclusion of the comma in the second file. Essentially the hash algorithms output string is a unique digital fingerprint of a file. If even a single bit is modified, the fingerprint will be completely different. This can be used to conclude that the files have changed.

Lab 7.1.1 Creating Hashes

There are numerous tools available to create a hash of a file; in Linux we can use the terminal and the command:

`md5sum filename`

Windows cannot do this by default and, therefore, requires a third party tool, such as HashCalc, which can be accessed from the *lab07 resources folder*.

Several files are available for you to download from the 'Hash Values' folder in the *lab07 resources folder*; there are doc files and JPG files. Some files look similar, but have slight, hard-to-notice changes. Using HashCalc or another tool, create an MD5 hash of each of the files.

Note the hashes and the filename.

Based on the above computed hashes, can you tell which of the files are exactly the same?

Lab 7.1.2 Seeing the Changes

Now, perform the following tasks :

Create a new text document.

In this file, type in **"Fundamentals of Digital Forensics AED 10.00"**

Save this file on your Desktop as **price.txt**, and close it.

Calculate the MD5 hash of this file. [Record this.](#)

Now make a copy of the file **"price.txt"** and call it **"copy of price.txt"**. **DO NOT MODIFY** or **OPEN** this file.

Calculate the MD5 hash of this file. [Record this.](#)

Now compare the hash values. [What can you learn from the hash of these two files? Why?](#)

Next, open the file **price.txt** you created earlier.

Change the statement **"Fundamentals of Digital Forensics AED 10.00"** to **"Fundamentals of Information Security AED 1000"**

Save the file (over-write it)

Just as before, calculate the hash of this edited file.

Compare the hash values to the ones you found before. [What can you see and why?](#)

Lab 7.2: Digital Forensic Imaging and analysis

The goal of digital forensics is to perform crime investigations by using evidence from digital data to find who was responsible for a particular crime, when and where.

Many digital forensic investigation toolkits have been developed over the past decades to facilitate the investigation and help digital forensic practitioners. Police departments and investigation agencies select the tools based on various factors including budget and available experts on the team, as well as the type of cases they are dealing with.

In this lab you will be introduced to two well-known forensic investigation toolkits.

Lab 7.2.1: Autopsy

[Autopsy](#) is a digital forensics platform and graphical interface to the [Sleuth Kit](#) and other digital forensics tools. It is used by law enforcement, military, and corporate examiners to investigate what happened on a computer. Autopsy has different analysis features such as:

- Timeline Analysis - Advanced graphical event viewing interface
- Hash Filtering - Flag known bad files and ignore known good.
- [Keyword Search](#) - Indexed keyword search to find files that mention relevant terms.
- [Web Artifacts](#) - Extract history, bookmarks, and cookies from Firefox, Chrome, and IE.
- Data Carving - Recover deleted files from unallocated space using [PhotoRec](#)
- Multimedia - Extract EXIF from pictures and watch videos.
- Indicators of Compromise - Scan a computer using STIX.

Download and install Autopsy, if you do not already have it installed on your machine.

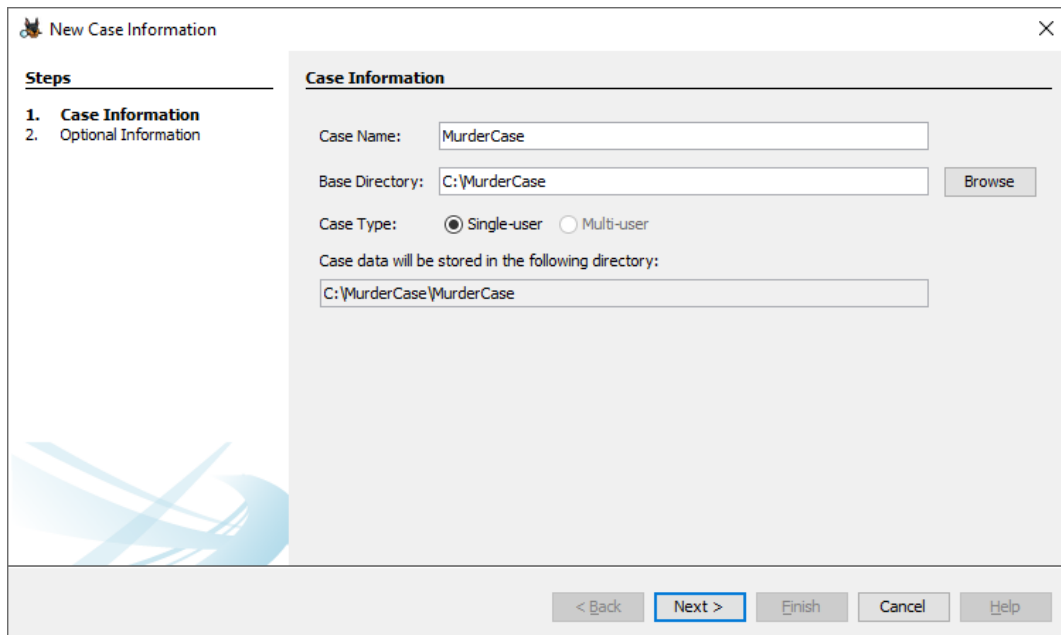
<https://www.autopsy.com/download/>

In your working directory, create a new folder and call it “Murdercase”. Save the Murder image file (downloaded from *lab07 resources* on Canvas) to the “Murdercase” folder. Open the Autopsy application. You should see a window as shown in Figure 1.



Figure1. Autopsy create a case

Click on “Create New Case”, a window will appear, as shown in Figure 2.



New Case Information

Steps

1. Case Information
2. Optional Information

Case Information

Case Name:

Base Directory:

Case Type: ☒ Single-user ☐ Multi-user

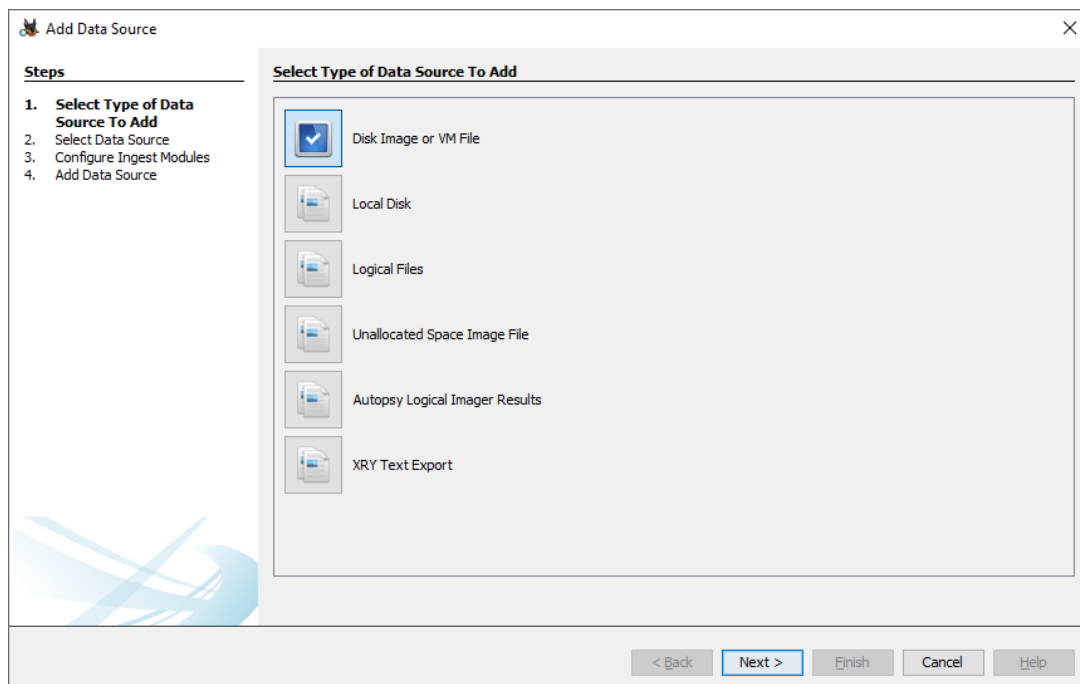
Case data will be stored in the following directory:

< Back **Next >** Finish Cancel Help

Figure 2. Add case information

In case Name, add “MurderCase”, browse to the “Murdercase” folder in your working directory. Click “Next” and enter Case number as “1” and your name as *Examiner*. Click Finish.

Another window will appear, as shown in Figure 3: Choose disk image or VM file.



Add Data Source

Steps

1. Select Type of Data Source To Add
2. Select Data Source
3. Configure Ingest Modules
4. Add Data Source

Select Type of Data Source To Add

- ☒ Disk Image or VM File
- ☐ Local Disk
- ☐ Logical Files
- ☐ Unallocated Space Image File
- ☐ Autopsy Logical Imager Results
- ☐ XRY Text Export

< Back **Next >** Finish Cancel Help

Figure 3. Choose the source of evidence

Browse to the “Murdercase” folder and choose Murder.E01, as shown in Figure 4.

Add Data Source

Steps

1. Select Type of Data Source To Add
- 2. Select Data Source**
3. Configure Ingest Modules
4. Add Data Source

Select Data Source

Path:

☐ Ignore orphan files in FAT file systems

Time zone:

Sector size:

Hash Values (optional):

MD5:

SHA-1:

SHA-256:

NOTE: These values will not be validated when the data source is added.

< Back Next > Finish Cancel Help

Figure 4. Select data source

Click “Next” to Configure Module window as shown in Figure 5. Untick Keyword Search, Virtual Machine Extractor, iOS Analyzer (if present) and Android Analyzer.

Add Data Source

Steps

1. Select Type of Data Source To Add
2. Select Data Source
- 3. Configure Ingest Modules**
4. Add Data Source

Configure Ingest Modules

Run ingest modules on:

☒ File Type Identification
☒ Extension Mismatch Detector
☒ Embedded File Extractor
☒ Exif Parser
☐ Keyword Search
☒ Email Parser
☒ Encryption Detection
☒ Interesting Files Identifier
☒ Correlation Engine
☒ PhotoRec Carver
☐ Virtual Machine Extractor
☒ Data Source Integrity
☐ Plaso
☐ **Android Analyzer**

Select All Deselect All History

The selected module has no per-run settings.

Extracts Android system and third-party app data.

Global Settings

< Back **Next >** Finish Cancel Help

Figure 5. Configure processing modules

Click Next and Finish.

Note the analysing status bar in the bottom right hand corner (see Figure 6).

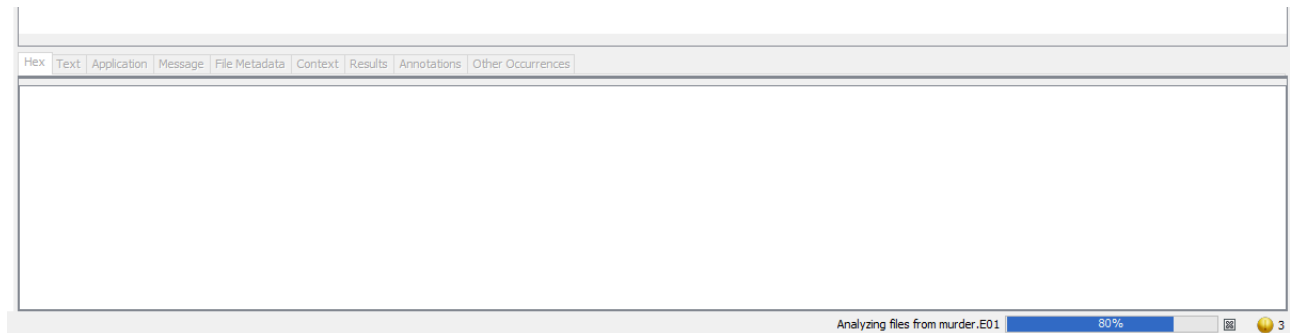


Figure 6. Analysing bar

Read the following Scenario, while the case processing is getting completed.

CASE REFERENCE: CSL/0294/10

TYPE: Murder

OIC: DC R SMYTH

SUMMARY:

Steven Turner was arrested on 29th October 2010 under suspicion of rape and murder of Jane Simmons, a prostitute well known to the Police around London Rd of Leicester.

The body was found on the 26th of October in a hotel room with shotgun wounds by the cleaner at around 10AM. The time of death is believed to have been between 10.30pm on Sunday 24th and 1am on Monday 25th October 2010.

At the time of Turner's arrest, he was found to be carrying a USB device and a mobile phone. The USB stick was imaged by a colleague and it is called "Murder.E01".

TASK: Analyse the image using Autopsy and identify any information related to the Murder of Jane Simmons.

SUSPECT(S):

NAME: Steven Turner

D.O.B: 27/4/75

OCCUPATION: Unemployed labourer

VICTIM(S):

NAME: Jane Simmons

D.O.B: 6/8/81

OCCUPATION: Sex worker

After the case processing completed, you should see a window similar to the screenshot shown in Figure 7 :

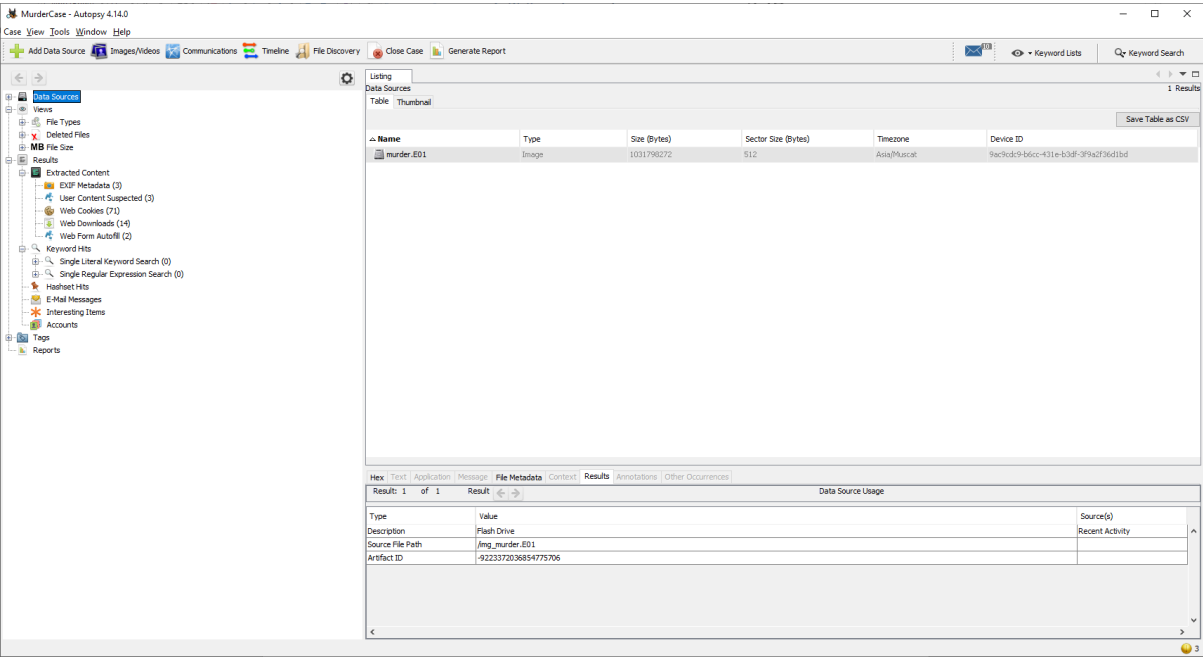


Figure 7. Processed case in Autopsy

What is the MD5 hash for the “Murder.E01” image?

Identify Device ID and the timeZone:

Click on Murder.E01, to expand it. What does the red X mean beside the Documents folder?

What sort of evidence can you identify for Murder.E01 by using Autopsy? Report your findings.

(Hint: Click on Murder.E01 look inside Documents, any related images or Documents. Also look at the visited sites). Record metadata details related to the evidence you found including MD5 hash for each evidence file.

Lab 7.2.2: FTK (Forensic Toolkit)

In Lab02, you were introduced to the FTK forensic investigation tool. Create a new folder called 'FTK lab07 test'. You can use the FTK Imager version that you used as part of Lab 02 on Canvas (if you do not have FTK installed, refer to lab 02 sheet for instructions on FTK installation).

Repeat the same process to perform the forensic analysis in FTK using the same image as in lab 7.2.1 (Murder.E01). If you use FTK imager, you will not be able to create a new case, you can skip creating a new case.

Take a look at the different overview views FTK can offer to view files. Do you believe it helps with analysis? Why is this useful for analysis?

What is the total number of files processed in this evidence file (Murder.E01) by FTK?

How many deleted files?

How many graphics files?

MAC Times

MAC (Modified, Accessed, creation) times are pieces of file system metadata which record when certain events pertaining to a computer file occurred most recently. In the example image file, look at some MAC Times for files in the Graphics folder.

Do you notice anything? Do some of the graphics files have their modified date before creation and accessed dates? If yes, why do you think that happened? If all the dates correspond, you can write no as an answer.

End of Lab Sheet